which is stuck at argon fluoride (ArF) source with a wavelength of 193nm. Optical advancements like drenching has made it possible to fabricate such circuits.

There were likewise different developments en route such as high-K metal gate that reduced the leakage power issue to a certain extent. Power supply voltage keeps on downsizing, fundamentally saving money on unique and static power. Short channel effects are diminished altogether lowering leaker power, drain tunneling, etc.

Table 1

Truth Table for 2 to 4 Decoder

A	B	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

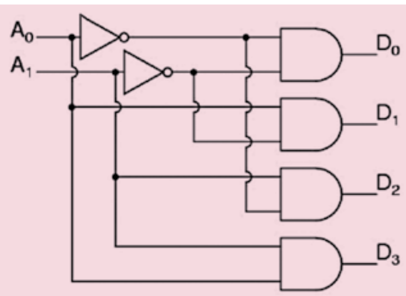

Fig. 2: Block diagram of 2 to 4 decoder

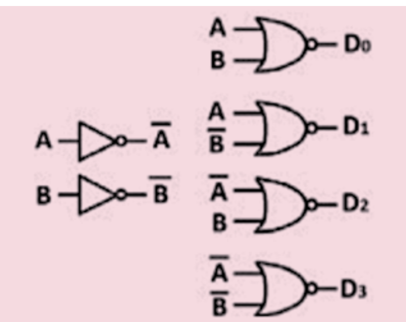

Fig. 3: 2 to 4 decoder using 20 CMOS transistors

Every single chip produced today uses very large scale integration (VLSI) techniques and has a great many gates and billions of individual transistors of miniscule size. There is a demand to decrease their size and power requirement but increase effectiveness. But power dissipation has turned out to be an essential parameter in both simple and advanced circuits. Dynamic power, clock power, and normal leakage power of the chips are diminished by up to 55%, 29%, and 53%, respectively while maintaining comparative speed with respect to the circuits executed with CMOS and CNTFET technology.

CNTFET

Carbon nanotube field effect transistors (CNTFETs) make up high-performance, low-power chips. A CNTFET is an FET that uses a single or many carbon nanotubes (CNTs) as the channel material as opposed to mass silicon in the standard MOSFET structure. CNTFETs were made from single CNTs (joined by laser expulsion) by oxidising silicon wafers that had been pre-patterned with gold or platinum terminals.

CNTs were introduced in 1991 by Japanese physicist S. lijima. CNT is a nanoscale tube that is made up of rolled sheets of graphene (see Fig. 1). It can be single-walled (SWCNT) or multi-walled (MWCNT). The SWCNT comprises a single nanotube, while MWCNT comprises numerous nanotubes with interlayer spacing of 0.34nm.

SWCNTs show properties such as semi-ballistic carrier transport and prevent short channel effects. These can be either semiconducting or metallic in nature. SWCNT is addressed by a whole number pair (n, m) called chirality vector. The nanotube is metallic if $n = m$, or $n - m = 3i$, where 'i' is a number; in any case, the SWCNT is semiconducting.

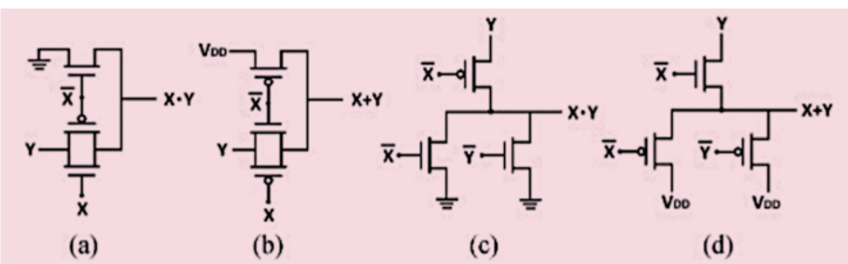

Fig. 4: AND/OR gates: (a) TGL AND gate, (b) TGL OR gate, (c) DVL AND gate, (d) DVL OR gate

Overview of 2 to 4 decoders

A decoder is a combinational circuit that converts or decodes the n inputs to 2^n outputs. It is basically a multi-input, multi-output logic circuit that converts the input to output codes—keeping the input and output codes different. The input code mostly has lesser number of bits than the output. Decoder performs the function reverse to that of an encoder. There can be many types of decoders depending on the number of inputs available such as 2 to 4 decoder, or 3 to 8 decoder.

In 2:4 decoder there are 2 inputs and 4 outputs as can be seen in Table 1. In each of the cases any one of the outputs is set to 1 while others are set to 0. A transforming 2-4 decoder produces the reciprocal minters I0-I3, and accordingly the chosen yield is set to 0 and the rest are set to 1.

The 2 to 4 decoder (see Fig. 2) has two inputs A1 and A0 and four outputs D3, D2, D1, and D0. As a decoder, this circuit takes an n-bit parallel number and produces an output on one of the 2^n output lines. The decoder comprises four AND gates and two inverters.

Using CMOS technology. While using CMOS technology, NAND and NOR gates are preferred rather than AND or OR gates as the latter require six transistors while the former require only four. This reduces the transistor count as well as overall efficiency. A 2 to 4 decoder can be implemented with four NOR gates and two inverters (see Fig. 3), that is, a total of twenty transistors.

Using mixed logic. Transmission gates can be easily used to design the AND/OR gates as shown in Fig. 4, thus making the decoders efficiently. The mixed logic uses transmission gates and 'pass' transistors to implement a circuit. Pass transistor logic (PTL) was introduced in the 1990's when many other designs were available in the market. It provides a suitable alternative to CMOS logic, keeping in mind the goals to improve factors such as speed, power, and area. The difference in such circuits is that the